



IT-SIMPLERA INSTITUTE

Cyber Security Analyst Internship Program

WEEK 03 ASSIGNMENT

Cybersecurity Fundamentals & Security Concepts

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1. Introduction

This report documents Week 03 of the Cyber Security Analyst Internship at IT-Simplera Institute. The assignment builds foundational networking and reconnaissance skills that underpin day-to-day analyst work: knowing which protocols and ports run on a network, telling secure from insecure communication, using Kali Linux tools for connectivity testing and service discovery, understanding how DNS and the browser-to-server handshake work end to end, and using OSINT tools such as whois and whatweb to a target's public footprint.

2. Task 1 — Protocols and Port Numbers

The table below lists 15 common network protocols with their default port, transport type, purpose, and whether they are secure or insecure by default.

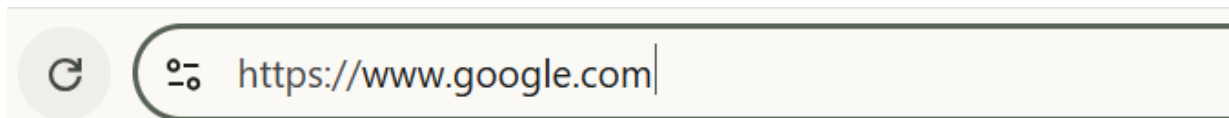
Protocol	Port	TCP/UDP	Purpose	Secure / Insecure
HTTP	80	TCP	Transfers unencrypted web pages between browser and server	Insecure
HTTPS	443	TCP	Transfers web pages encrypted using TLS/SSL	Secure
FTP	20/21	TCP	Transfers files between client and server	Insecure
SFTP	22	TCP	Transfers files securely over SSH	Secure
SSH	22	TCP	Secure remote login and command execution	Secure
Telnet	23	TCP	Remote login to a device (plain text)	Insecure
SMTP	25	TCP	Sends email between mail servers	Insecure
DNS	53	TCP/UDP	Resolves domain names to IP addresses	Insecure
DHCP	67/68	UDP	Automatically assigns IP addresses to devices	Insecure
TFTP	69	UDP	Simple, connectionless file transfer	Insecure
POP3	110	TCP	Downloads email from a mail server	Insecure
IMAP	143	TCP	Retrieves and manages email on the server	Insecure
SNMP	161/162	UDP	Monitors and manages network devices	Insecure
LDAP	389	TCP	Accesses and maintains directory services	Insecure

RDP	3389	TCP	Remote desktop access to Windows systems	Insecure by default
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3. Task 2 — HTTP vs HTTPS Analysis

Ten widely used websites were checked for their default protocol and HTTP→HTTPS redirect behavior. As of 2026, virtually all major sites enforce HTTPS and reject plain HTTP via HSTS (HTTP Strict Transport Security), which instructs the browser to never attempt an unencrypted connection again after the first secure visit.

Website-01 (Google.com)



Type: Https

HTTP → HTTPS Redirect? YES

Is the connection secure?

Yes. Google uses HTTPS with a valid SSL/TLS certificate to encrypt data between your browser and its servers. This protects user information from interception and tampering.

Website-02 (Facebook.com)



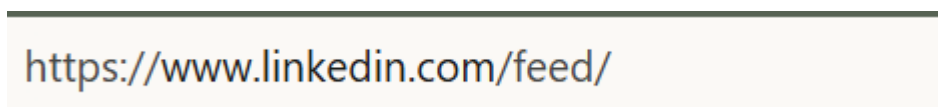
Type: Https

HTTP → HTTPS Redirect? YES

Is the connection secure?

Yes. Facebook forces secure HTTPS connections and uses TLS encryption. This helps protect logins, messages, and other sensitive data.

Website-03 (LinkedIn.com)



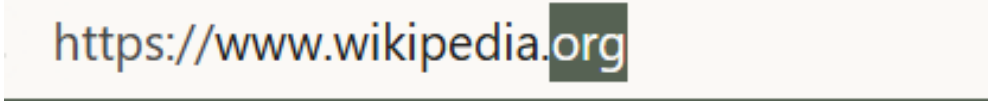
Type: Https

HTTP → HTTPS Redirect? YES

Is the connection secure?

Yes. LinkedIn automatically redirects users to HTTPS and uses TLS certificates. This ensures secure communication and protects personal and professional information.

Website-04 (Wikipedia.org)



https://www.wikipedia.org

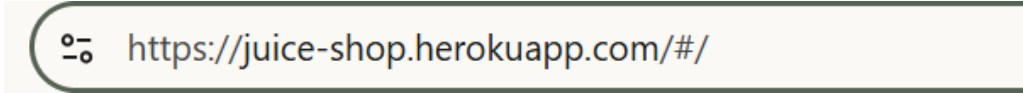
Type: Https

HTTP → HTTPS Redirect? YES

Is the connection secure?

Yes. Wikipedia uses HTTPS by default with valid TLS certificates. This prevents attackers from modifying or spying on the content you access.

Website-05 (juice-shop.herokuapp.com)



https://juice-shop.herokuapp.com/#/

Type: Https

HTTP → HTTPS Redirect? YES

Is the connection secure?

Yes. Although OWASP Juice Shop is intentionally vulnerable at the application level, the network connection itself is protected by HTTPS/TLS. The vulnerabilities are part of the web application, not the transport encryption.

Website-06 (github.com)



https://github.com|

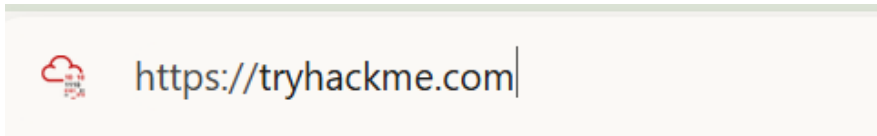
Type: Https

HTTP → HTTPS Redirect? YES

Is the connection secure?

Yes. GitHub enforces HTTPS using strong TLS encryption. This secures source code, authentication, and other user data during transmission.

Website-07 (tryhackme.com)



Type: Https

HTTP → HTTPS Redirect? YES

Is the connection secure?

Yes. TryHackMe uses HTTPS with TLS encryption and automatically redirects HTTP traffic. This protects user accounts, labs, and learning activities.

Website-08 (microsoft.com)



Type: Https

HTTP → HTTPS Redirect? YES

Is the connection secure?

Yes. Microsoft uses HTTPS with trusted SSL/TLS certificates and redirects HTTP requests to HTTPS. This ensures confidentiality and integrity of the connection.

Website-09(Geeksforgeeks):



Type: Https

HTTP → HTTPS Redirect? YES

Is the connection secure?

Yes. GeeksforGeeks serves its website over HTTPS and redirects HTTP requests. Data exchanged between the browser and server is encrypted.

Website-10(Forever.com):



Type: Http

HTTP → HTTPS Redirect? No

Is the connection secure?

No. The website only supports HTTP and does not provide encryption. Data sent between the browser and the server can be intercepted or modified by attackers, so the connection is not secure.

How to verify this yourself

- Open each site with `http://` explicitly typed (not `https://`) and watch whether the browser address bar flips to `https://` automatically.
- In Chrome/Firefox dev tools (Network tab), check the first request's status code a 301/308 redirect to an `https://` URL confirms enforced HTTPS.
- Click the padlock icon to inspect the TLS certificate issuer and validity dates.

4. Task 3 — Kali Linux Network Reconnaissance

4.1 Connectivity test

Command:

To check the connectivity we always use ping command.

```
(kali@kali)-[~]
$ ping google.com
PING google.com (142.250.202.142) 56(84) bytes of data:
64 bytes from lbg02s04-in-f14.1e100.net (142.250.202.142): icmp_seq=1 ttl=255 time=61.1 ms
64 bytes from lbg02s04-in-f14.1e100.net (142.250.202.142): icmp_seq=2 ttl=255 time=70.7 ms
64 bytes from lbg02s04-in-f14.1e100.net (142.250.202.142): icmp_seq=3 ttl=255 time=62.2 ms
64 bytes from lbg02s04-in-f14.1e100.net (142.250.202.142): icmp_seq=4 ttl=255 time=53.5 ms
64 bytes from lbg02s04-in-f14.1e100.net (142.250.202.142): icmp_seq=5 ttl=255 time=69.0 ms
64 bytes from lbg02s04-in-f14.1e100.net (142.250.202.142): icmp_seq=6 ttl=255 time=110 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5009ms
rtt min/avg/max/mdev = 53.518/71.051/109.818/18.222 ms
```

4.2 Service discovery nmap

Command:

1. Nmap Scanme.nmap.org:

```
(kali@kali)-[~]
$ nmap Scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-18 12:45 EDT
nmap scan report for Scanme.nmap.org (45.33.32.156)
Host is up (0.0027s latency).
Other addresses for Scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
DNS record for 45.33.32.156: scanme.nmap.org
All 1000 scanned ports on Scanme.nmap.org (45.33.32.156) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 4.93 seconds
```

2. Nmap Scan (Local IP)

```
(kali@kali)-[~]
$ nmap
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-18 12:40 EDT
Nmap scan report for 
Host is up (0.0000040s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
```

3. Bonus Scan

```
(kali@kali)-[~]
$ nmap -sV 
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-18 12:41 EDT
Nmap scan report for 
Host is up (0.0000040s latency).
```

```
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
```

```
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds
```

5. Task 4 — What Happens When You Type a URL and Press Enter?

Typing a URL and pressing Enter triggers a fast, layered sequence of events across the network stack. The full path is summarized below.

5.1 DNS Lookup

The browser first checks its own cache, then the OS cache and hosts file, for the domain name. If not found, it queries a configured DNS resolver (often the ISP's or a public resolver like 1.1.1.1 or 8.8.8.8), which recursively queries root, TLD, and authoritative name servers until it returns the IP address mapped to the domain.

5.2 IP Address Resolution

Once resolved, the browser now has a destination IP address (IPv4 or IPv6) to connect to. This result is cached locally for a period defined by the DNS record's TTL, so repeat visits skip the full lookup.

5.3 TCP Three-Way Handshake

The browser opens a TCP connection to the server on the destination port (80 for HTTP, 443 for HTTPS) using the SYN, SYN-ACK, ACK exchange: the client sends SYN, the server replies to SYN-ACK, and the client confirms with ACK, establishing a reliable, ordered connection.

5.4 TLS/HTTPS Handshake

For HTTPS, a TLS handshake follows immediately: the client and server negotiate a TLS version and cipher suite, the server presents its certificate (verified against a trusted Certificate Authority), and both sides derive a shared session key. From this point, all data exchanged is encrypted.

5.5 HTTP Request

The browser sends an HTTP request (e.g. GET /) over the now-secure connection, including headers such as the requested path, cookies, user-agent string, and accepted content types.

5.6 Server Response

The web server processes the request often involving an application server, database queries, or a CDN cache lookup and returns an HTTP response with a status code (200 OK, 301 redirect, 404 not found, etc.), response headers, and the requested content (HTML, JSON, images, etc.).

5.7 Browser Rendering the Webpage

The browser parses the returned HTML into a DOM tree, fetches linked resources (CSS, JavaScript, images, fonts) often triggering additional DNS/TCP/TLS cycles for other domains builds the CSS render tree, executes JavaScript, and paints the final page to the screen. Modern browsers also run security checks throughout, such as validating the TLS certificate chain and applying Content Security Policy rules.

6. Task 5 — Networking Cheat Sheet

A quick-reference cheat sheet covering the same 15 protocols, now including the specific security risk each one carries and a secure alternative where one exists.

Protocol	Port	TCP/UDP	Purpose	Security Risk	Secure Alternative
HTTP	80	TCP	Web page delivery	Data sent in plain text; sniffable	HTTPS (443)
HTTPS	443	TCP	Encrypted web page delivery	Weak TLS config, expired certs	N/A
FTP	20/21	TCP	File transfer	Credentials and files sent in clear text	SFTP / FTPS
SFTP	22	TCP	Secure file transfer over SSH	Weak SSH keys, brute force	N/A
SSH	22	TCP	Encrypted remote administration	Default creds, brute force, outdated ciphers	N/A
Telnet	23	TCP	Plain-text remote login	Fully unencrypted; credentials exposed	SSH
SMTP	25	TCP	Mail relay between servers	Open relay abuse, spoofing	SMTPS (465/587)
DNS	53	TCP/UDP	Domain name resolution	Cache poisoning, spoofing, amplification DDoS	DNSSEC / DoH / DoT
DHCP	67/68	UDP	Automatic IP address assignment	Rogue DHCP server, IP exhaustion	DHCP snooping

TFTP	69	UDP	Lightweight file transfer (no auth)	No authentication at all	SFTP
POP3	110	TCP	Download mail from server	Credentials sent in clear text	POP3S (995)
IMAP	143	TCP	Sync mail with server	Credentials sent in clear text	IMAPS (993)
SNMP	161/162	UDP	Device monitoring/management	Default community strings ('public')	SNMPv3
LDAP	389	TCP	Directory service queries	Clear-text bind credentials	LDAPS (636)
RDP	3389	TCP	Remote desktop access	Brute force, BlueKeep-type exploits	VPN + NLA + MFA

7. Task 6 — Kali Linux Web Enumeration

WhatWeb

Function: WhatWeb is a web technology fingerprinting tool used to identify the technologies running on a website.

It can detect:

- Web server (Apache, Nginx, IIS)
- Content Management System (WordPress, Joomla, Drupal)
- Programming languages (PHP, ASP.NET, etc.)
- JavaScript libraries
- Frameworks
- Analytics tools
- Security headers and other technologies

Analyze Google.com:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali) ~
$ whatweb google.com
http://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[gws], IP[142.250.202.142], RedirectLocation[http://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only, X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]]
https://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[gws], IP[142.250.202.110], RedirectLocation[https://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only, alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
http://www.google.com/ [200 OK] Cookies[AEC,NID, __Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[142.251.155.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only, X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]]
https://www.google.com/ [200 OK] Cookies[AEC,NID, __Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[142.251.157.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only, accept-ch, alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
```

Analyze Github.com:

```

(kali@kali)-[~]
└─$ whatweb github.com
http://github.com [301 Moved Permanently] Country[UNITED STATES][US], IP[20.207.73.82], RedirectLocation[https://github.com/]
https://github.com [200 OK] Content-Language[en-US], Cookies[_gh_sess,octo,logged_in], Country[UNITED STATES][US], Email[you@domain.com], HTML5, HTTPServer[github.com], HttpOnly[_gh_sess,logged_in], IP[20.207.73.82], Open-Graph-Protocol[object][1401488693436528], OpenSearch[/opensearch.xml], Script[application/javascript,module], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub · Change is constant. GitHub keeps you ahead. · GitHub], UncommonHeaders[x-content-type-options,referrer-policy,content-security-policy,x-github-request-id], X-Frame-Options[deny], X-XSS-Protection[0]
https://github.com/ [200 OK] Content-Language[en-US], Cookies[_gh_sess,octo,logged_in], Country[UNITED STATES][US], Email[you@domain.com], HTML5, HTTPServer[github.com], HttpOnly[_gh_sess,logged_in], IP[20.207.73.82], Open-Graph-Protocol[object][1401488693436528], OpenSearch[/opensearch.xml], Script[application/javascript,module], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub · Change is constant. GitHub keeps you ahead. · GitHub], UncommonHeaders[x-content-type-options,referrer-policy,content-security-policy,x-github-request-id], X-Frame-Options[deny], X-XSS-Protection[0]

```

WHOIS

Function:

WHOIS is a protocol/tool used to retrieve domain registration information about a website.

It provides details such as:

- Domain owner/registrar (if not privacy-protected)
- Domain registration date
- Expiration date
- Name servers
- Registrar information
- Domain status

Analyze google.com:

```

(kali@kali)-[~]
$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-18T16:53:34Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,

```

```

Registrant Country: US
Registrant Email: Select Request Email Form at https://domains.markmonitor.com/whois/google.com
Tech Email: Select Request Email Form at https://domains.markmonitor.com/whois/google.com
Name Server: ns4.google.com
Name Server: ns2.google.com
Name Server: ns3.google.com
Name Server: ns1.google.com
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2026-07-18T16:50:37+0000 <<<

```

For more information on WHOIS status codes, please visit:
<https://www.icann.org/resources/pages/epp-status-codes>

If you wish to contact this domain's Registrant or Technical contact, and such email address is not visible above, you may do so via our web form, pursuant to ICANN's Temporary Specification. To verify that you are not a robot, please enter your email address to receive a link to a page that facilitates email communication with the relevant contact(s).

Web-based WHOIS:
<https://domains.markmonitor.com/whois/contact/google.com>

If you have a legitimate interest in viewing the non-public WHOIS details, send your request and the reasons for your request to whoisrequest@markmonitor.com and specify the domain name in the subject line. We will review that request and may ask for supporting documentation and explanation.

The data in MarkMonitor's WHOIS database is provided for information purposes, and to assist persons in obtaining information about or related to a domain name's registration record. While MarkMonitor believes the data to be accurate, the data is provided "as is" with no guarantee or warranties regarding its accuracy.

By submitting a WHOIS query, you agree that you will use this data only for lawful purposes and that, under no circumstances will you use this data to:

- (1) allow, enable, or otherwise support the transmission by email, telephone, or facsimile of mass, unsolicited, commercial advertising, or spam; or
- (2) enable high volume, automated, or electronic processes that send queries, data, or email to MarkMonitor (or its systems) or the domain name contacts (or its systems).

MarkMonitor reserves the right to modify these terms at any time.

By submitting this query, you agree to abide by this policy.

MarkMonitor Domain Management(TM)
 Protecting companies and consumers in a digital world.

Analyze github.com

```

(kali@kali)-[~]
$ whois github.com
Domain Name: GITHUB.COM
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-09T09:16:32Z
Creation Date: 2007-10-09T18:20:50Z
Registry Expiry Date: 2026-10-09T18:20:50Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: DNS1.P08.NSONE.NET
Name Server: DNS2.P08.NSONE.NET
Name Server: DNS3.P08.NSONE.NET
Name Server: DNS4.P08.NSONE.NET
Name Server: NS-1283.AWSDNS-32.ORG
Name Server: NS-1707.AWSDNS-21.CO.UK
Name Server: NS-421.AWSDNS-52.COM
Name Server: NS-520.AWSDNS-01.NET
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-18T16:54:49Z <<<

```

For more information on Whois status codes, please visit <https://icann.org/epp>

NOTICE: The expiration date displayed in this record is the date the registrar's sponsorship of the domain name registration in the registry is currently set to expire. This date does not necessarily reflect the expiration date of the domain name registrant's agreement with the sponsoring registrar. Users may consult the sponsoring registrar's Whois database to view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois database through the use of electronic processes that are high-volume and automated except as reasonably necessary to register domain names or modify existing registrations; the Data in VeriSign Global Registry Services' ("VeriSign") Whois database is provided by VeriSign for information purposes only, and to assist persons in obtaining information about or related to a domain name registration record. VeriSign does not guarantee its accuracy. By submitting a Whois query, you agree to abide by the following terms of use: You agree that you may use this Data only for lawful purposes and that under no circumstances will you use this Data to: (1) allow, enable, or otherwise support the transmission of mass unsolicited, commercial advertising or solicitations via e-mail, telephone,

```

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: github.com
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-07T09:16:33+0000
Creation Date: 2007-10-09T18:20:50+0000
Registrar Registration Expiration Date: 2026-10-09T00:00:00+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact: https://corp.markmonitor.com/domain/ui/abuse-report
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibited)
Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProhibited)
Domain Status: clientDeleteProhibited (https://www.icann.org/epp#clientDeleteProhibited)
Registrant Organization: GitHub, Inc.
Registrant Country: US
Registrant Email: Select Request Email Form at https://domains.markmonitor.com/whois/github.com
Tech Email: Select Request Email Form at https://domains.markmonitor.com/whois/github.com
Name Server: dns2.p08.nsone.net
Name Server: dns3.p08.nsone.net
Name Server: ns-421.awsdns-52.com
Name Server: ns-520.awsdns-01.net
Name Server: ns-1707.awsdns-21.co.uk
Name Server: dns1.p08.nsone.net
Name Server: dns4.p08.nsone.net
Name Server: ns-1283.awsdns-32.org
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2026-07-18T16:54:27+0000 <<<

For more information on WHOIS status codes, please visit:
https://www.icann.org/resources/pages/epp-status-codes

If you wish to contact this domain's Registrant or Technical
contact, and such email address is not visible above, you may do so via our web
form, pursuant to ICANN's Temporary Specification. To verify that you are not a
robot, please enter your email address to receive a link to a page that
facilitates email communication with the relevant contact(s).

Web-based WHOIS:
https://domains.markmonitor.com/whois/contact/github.com

If you have a legitimate interest in viewing the non-public WHOIS details, send
your request and the reasons for your request to whoisrequest@markmonitor.com
and specify the domain name in the subject line. We will review that request and
may ask for supporting documentation and explanation.

```

1. Google.com

Web Server

- Google Web Server (GWS)
- HTTPS enabled
- HTML5
- Cookies
- X-Frame-Options
- X-XSS-Protection

Domain Registrar

- MarkMonitor Inc.

Registration Date

- 15 September 1997

Name Servers

- ns1.google.com

- ns2.google.com
- ns3.google.com
- ns4.google.com

Interesting Information Discovered

- Automatically redirects HTTP requests to HTTPS.
- Uses Google's own web server (GWS).
- Protected by domain security lock statuses.
- Communication is encrypted using HTTPS

2. GitHub.com**Web Server / Technologies Detected**

- HTTPS
- HTML5
- Open Graph Protocol
- Strict-Transport-Security (HSTS)
- Cookies
- X-Frame-Options
- X-XSS-Protection

Domain Registrar

- MarkMonitor Inc.

Registration Date

- 9 October 2007

Name Servers

- dns1.p08.nsone.net
- dns2.p08.nsone.net
- dns3.p08.nsone.net
- dns4.p08.nsone.net
- ns-421.awsdns-52.com
- ns-520.awsdns-01.net
- ns-1707.awsdns-21.co.uk
- ns-1283.awsdns-32.org

Interesting Information Discovered

- Automatically redirects HTTP to HTTPS.
- Uses HSTS to enforce secure HTTPS connections.
- Uses NS1 and AWS Route 53 for DNS services.
- Protected by domain security lock statuses.
- Implements multiple security headers to enhance website security.

9. Challenges Faced

- Distinguishing default-insecure protocols (e.g. SMTP, POP3, IMAP) from their secure counterparts required checking each protocol's secure variant and port separately, since the base protocol and its TLS-wrapped version share a name but not a port.
- Reconciling nmap scan results with “risk” required judgement rather than a fixed rule — an open port is only risky in context (unpatched service, weak auth, unnecessary exposure), which took extra research per finding.
- Running live reconnaissance tools (ping, nmap, whatweb) needs an actual Kali Linux environment with authorized network access; this had to be scoped separately from the desk research and cheat-sheet portions of the assignment.

10. Conclusion

This assignment reinforced the building blocks a cybersecurity analyst uses daily: recognizing protocols and their risk profile by port, verifying whether a site's transport is actually encrypted rather than assuming so, using reconnaissance tools to discover what a target exposes, and understanding the full request lifecycle from URL to rendered page. Together, the protocol table, HTTPS analysis, reconnaissance findings, URL-lifecycle explanation, cheat sheet, and network diagram form a practical reference that can be reused in future analyst work — particularly the cheat sheet, which condenses ports, risks, and secure alternatives into a single lookup table.

11. References

- *IANA Service Name and Transport Protocol Port Number Registry* iana.org/assignments/service-names-port-numbers
- *Nmap Reference Guide* nmap.org/book/man.html
- *Cloudflare Learning Center What happens when you enter a URL* cloudflare.com/learning
- *Mozilla Developer Network HTTP overview and TLS handshake* developer.mozilla.org
- *WHOIS public domain registry lookups for google.com and github.com* whois.com
- *Kali Linux Tools Documentation* kali.org/tools